

Multi-state models 3

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Competing risks and multi-state models course
May 5–8, 2026, Stockholm



Outline

Introduction

Multi-state models

- Theory
- Application
- Summary



Outline

Introduction

Multi-state models

Theory

Application

Summary

Dynamic prediction

- ▶ Want to answer questions like:
 - ▶ Given a Bone Marrow Transplant patient who has had acute Graft-versus-Host Disease after 60 days, and whose platelets have recovered after 120 days and who has had no further events after 1 year, what is then the probability of surviving relapse-free for 2 more years?
 - ▶ Given a breast cancer patient who is alive without local recurrence and distant metastasis at 1.5 years after surgery, what are the probabilities of remaining alive and event-free, of experiencing a local recurrence and/or distant metastasis, etc, for the coming 10 years?
- ▶ Common problem is to estimate the conditional probabilities of some clinical future events, given an (event) history, and possibly a set of values for prognostic factors of a patient



Two approaches

- ▶ Multi-state models
 - ▶ Probability calculations based on models for transition rates
 - ▶ Relies heavily on Markov assumption
 - ▶ Efficient if assumptions are met, computationally heavy
- ▶ Landmarking
 - ▶ Closer to the data
 - ▶ Can only be used for prediction, because it does not start from a model describing the process
 - ▶ Less efficient, robust, computationally “easy”

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Transition probabilities

- ▶ Prediction time s (after transplant, surgery, diagnosis)
- ▶ Suppose we are in state k at time s
- ▶ What is the probability that we are in state ℓ at time t ?
- ▶ The **transition probability**

$$P_{k\ell}(s, t) = P(X(t) = \ell \mid X(s) = k)$$

- ▶ Gather these transition probabilities in an $S \times S$ matrix

$$\mathbf{P}(s, t) = \{P_{k\ell}(s, t)\}$$

- ▶ We discussed the Aalen-Johansen estimator

$$\hat{\mathbf{P}}(s, t) = \prod_{s < u \leq t} (\mathbf{I} + d\hat{\mathbf{H}}(u))$$

with $\hat{\mathbf{H}}(u) = \{\hat{H}_{k\ell}(u)\}$ a matrix with all estimated transition hazards and on the diagonal $\hat{H}_{kk}(u) = -\sum_{\ell \neq k} \hat{H}_{k\ell}(u)$

Computational aspects

$$\hat{\mathbf{P}}(s, t) = \prod_{s < u \leq t} (\mathbf{I} + d\hat{\mathbf{H}}(u))$$

Forward prediction

- ▶ If we fix s and vary t , then $\hat{\mathbf{P}}(s, t)$ changes value only at the event times $s < u_1 < u_2 < \dots$
 - ▶ $\hat{\mathbf{P}}(s, u_1), \hat{\mathbf{P}}(s, u_2), \dots$ may be calculated by successive $\mathbf{I} + d\hat{\mathbf{H}}(u)$ matrix multiplications, forward in time

Fixed horizon prediction

- ▶ If we fix a horizon t and vary s , then $\hat{\mathbf{P}}(s, t)$ changes value only at event times $t > u_1 > u_2 > \dots$
 - ▶ $\hat{\mathbf{P}}(u_1, t), \hat{\mathbf{P}}(u_2, t), \dots$ may be calculated by successive $\mathbf{I} + d\hat{\mathbf{H}}(u)$ matrix multiplications, backward in time

Variances

Warning: heavy formulas ahead!

- ▶ Since $\mathbf{P}(S, t) = \prod_{(s, t]} (\mathbf{I} + d\mathbf{H}(u))$ is a (complicated perhaps) function of $\mathbf{H}$, we can use the functional delta method to express the variance of $\hat{\mathbf{P}}(s, t)$ in terms of the variance of $\hat{\mathbf{H}}(t)$
- ▶ The (Greenwood) formula becomes

$$\widehat{\text{var}}(\hat{\mathbf{P}}(s, t)) = \sum_{u \in (s, t]} \{ \hat{\mathbf{P}}(u, t)^{\top} \otimes \hat{\mathbf{P}}(s, u-) \} \widehat{\text{var}}(\Delta \hat{\mathbf{H}}(u)) \cdot \{ \hat{\mathbf{P}}(u, t) \otimes \hat{\mathbf{P}}(s, u-)^{\top} \}.$$


Recursion formulas

- ▶ The direct formula of $\widehat{\text{var}}(\widehat{\mathbf{P}}(s, t))$ is complicated enough
- ▶ It turns out there is a recursion formula

$$\begin{aligned}\widehat{\text{var}}(\widehat{\mathbf{P}}(s, t)) = & \\ & \{(I + \Delta\widehat{\mathbf{H}}(t))^{\top} \otimes I\} \widehat{\text{var}}(\widehat{\mathbf{P}}(s, t-)) \{(I + \Delta\widehat{\mathbf{H}}(t)) \otimes I\} \\ & + \{I \otimes \widehat{\mathbf{P}}(s, t-)\} \widehat{\text{var}}(\Delta\widehat{\mathbf{H}}(t)) \{I \otimes \widehat{\mathbf{P}}(s, t-)^{\top}\}, \quad (1)\end{aligned}$$

- ▶ Formula is downright ugly, but very convenient
- ▶ This formula is implemented in `mstate`
- ▶ Subtly different recursion formulas for Aalen and Greenwood variances and for forward and fixed horizon prediction
- ▶ The above formulas can be found in Andersen et al. (1993), the other ones in de Wreede et al. (2010)

Covariates

- ▶ Given a set of (expanded) covariate values $\mathbf{Z}^*$ of a patient,

$$\hat{H}_{k\ell}(u | \mathbf{Z}^*) = \hat{H}_{k\ell,0}(u) \exp(\hat{\beta}^\top \mathbf{Z}_{k\ell}^*)$$

and

$$\hat{\mathbf{P}}(s, t | \mathbf{Z}^*) = \prod_{s < u \leq t} \left(\mathbf{I} + d\hat{\mathbf{H}}(u | \mathbf{Z}^*) \right)$$

- ▶ These two steps done in `mstate`
 - ▶ `msfit` to obtain estimates of patient-specific transition hazards and (co)variances
 - ▶ `probtrans` to obtain estimates of the transition probabilities and (co)variances

Two patients

Two patients

- ▶ Two patients, same age (reference, 55)
- ▶ Patient A (good risk patient): in CR, HLA-id sib donor, other sex match
- ▶ Patient B (poor risk patient): SD/PD, other donor, female donor to male recipient

Procedure in `mstate`

1. Calculate patient-specific cumulative hazards for all transitions (`msfit`)
2. Use result from `msfit` to calculate transition probabilities (`probtrans`)

Applying *msfit* (1)

- ▶ Like in *survfit*, define a data frame containing the covariate values
- ▶ Easiest to find patient in the original data set (no worries with factor levels)

```
> whA <- which(msc11$SIBdonor=="HLA-id sib" & msc11$remstat=="CR" & ms
> patA <- msc11[rep(whA[1],3), c(11:13)]
> patA$age10 <- 0
> patA$trans <- patA$strata <- 1:3
> attr(patA, "trans") <- tmat
> patA
```

An object of class 'msdata'

Data:

	remstat	SIBdonor	sexmatch	age10	strata	trans
1	CR	HLA-id sib	other	0	1	1
1.1	CR	HLA-id sib	other	0	2	2
1.2	CR	HLA-id sib	other	0	3	3

Applying *msfit* (2)

► Expand the covariates

```
> patA <- expand.covs(patA, covs, longnames=FALSE)
```

```
> patA
```

An object of class 'msdata'

Data:

	remstat	SIBdonor	sexmatch	age10	strata	trans	age10.1	age10.2
1	CR	HLA-id sib	other	0	1	1	0	0
1.1	CR	HLA-id sib	other	0	2	2	0	0
1.2	CR	HLA-id sib	other	0	3	3	0	0
	age10.3	remstat1.1	remstat1.2	remstat1.3	remstat2.1	remstat2.2		
1	0	0	0	0	0	0		0
1.1	0	0	0	0	0	0		0
1.2	0	0	0	0	0	0		0
	remstat2.3	SIBdonor.1	SIBdonor.2	SIBdonor.3	sexmatch.1	sexmatch.2		
1	0	0	0	0	0	0		0
1.1	0	0	0	0	0	0		0
1.2	0	0	0	0	0	0		0
	sexmatch.3							
1	0							
1.1	0							
1.2	0							

Applying *msfit* (3)

► Finally, the call to *msfit*

```
> msfA <- msfit(cox0, patA, trans=tmat)
> plot(msfA, lwd=2, xlim=c(0,100), ylim=c(0,5))
> title("Good risk patient")
```

Proportional baseline hazards models

- Here there are two strata (transitions 1 and 2/3)

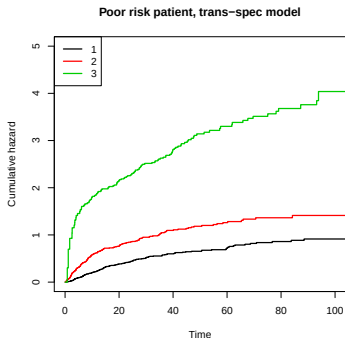
```
> patA_PBH <- patA
> patA_PBH$Z2.3 <- c(0, 0, 1)
> patA_PBH$strata <- c(1, 2, 2)
> msfA_PBH <- msfit(coxPBH, patA_PBH, trans=tmatrix)
> patA_PBH
```

An object of class 'msdata'

Data:

	remstat	SIBdonor	sexmatch	age10	strata	trans	age10.1	age10.2
1	CR	HLA-id	sib	other	0	1	1	0
1.1	CR	HLA-id	sib	other	0	2	2	0
1.2	CR	HLA-id	sib	other	0	2	3	0
	age10.3	remstat1.1	remstat1.2	remstat1.3	remstat2.1	remstat2.2		
1	0	0	0	0	0	0	0	0
1.1	0	0	0	0	0	0	0	0
1.2	0	0	0	0	0	0	0	0
	remstat2.3	SIBdonor.1	SIBdonor.2	SIBdonor.3	sexmatch.1	sexmatch.2		
1	0	0	0	0	0	0	0	0
1.1	0	0	0	0	0	0	0	0
1.2	0	0	0	0	0	0	0	0

-
- Figure 1 is a line graph titled "Good risk patient, trans-spec model". The y-axis is labeled "Cumulative hazard" and ranges from 0 to 5. The x-axis is labeled "Time" and ranges from 0 to 100. There are three curves plotted, labeled 1, 2, and 3 in the legend. Curve 1 (black) starts at (0,0) and increases slowly to approximately 0.2 at time 100. Curve 2 (red) starts at (0,0) and increases to approximately 0.4 at time 100. Curve 3 (green) starts at (0,0) and increases sharply to approximately 4.6 at time 100. All curves show a step-like pattern, indicating discrete events over time.



Applying *probtans*

- The result from *probtans* is a list, with item `[[k]]`, containing $P_{k,\ell}(s, t)$

```
> ptA <- probtans(msfA, predt=0, direction="forward")
> head(ptA[[1]], 12)
```

	time	pstate1	pstate2	pstate3	pstate4	se1
1	0.00000000	1.00000000	0.0000000000	0.0000000000	0	0.0000000000
2	0.03286771	0.9992737	0.0000000000	0.000726289	0	0.0007417846
3	0.23007395	0.9985457	0.0000000000	0.001454338	0	0.0010721138
4	0.26294166	0.9970885	0.0000000000	0.002911474	0	0.0015771811
5	0.29580937	0.9963574	0.0000000000	0.003642571	0	0.0017974026
6	0.32867707	0.9956251	0.0000000000	0.004374922	0	0.0020055866
7	0.46014790	0.9948920	0.0000000000	0.005107974	0	0.0022049319
8	0.49301561	0.9941584	0.0000000000	0.005841636	0	0.0023976497
9	0.52588332	0.9937067	0.0004516453	0.005841636	0	0.0024432958
10	0.59161873	0.9932545	0.0009038901	0.005841636	0	0.0024971700
11	0.62448644	0.9916134	0.0018091251	0.006577480	0	0.0027992990
12	0.65735415	0.9908750	0.0018091251	0.007315920	0	0.0029702652

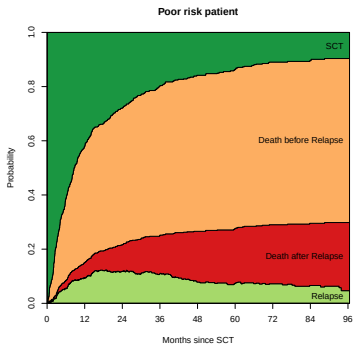
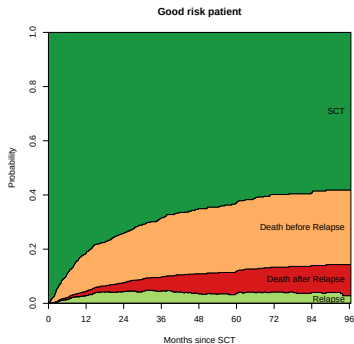
	se2	se3	se4
1	0.0000000000	0.0000000000	0
2	0.0000000000	0.0007417846	0
3	0.0000000000	0.0010721138	0



Plotting *probtrans*

```
> ptA <- probtrans(msfA, predt=0, direction="forward")
> plot(ptA, lwd=2, type="filled", ord=c(2,4,3,1), xlim=c(0,96.5), xlab=
> axis(1, at=seq(0,96,by=12), labels=seq(0,96,by=12))
> axis(2)
> title("Good risk patient")
> ptB <- probtrans(msfB, predt=0, direction="forward")
> plot(ptB, lwd=2, type="filled", ord=c(2,4,3,1), xlim=c(0,96.5), xlab=
> axis(1, at=seq(0,96,by=12), labels=seq(0,96,by=12))
> axis(2)
> title("Poor risk patient")
```

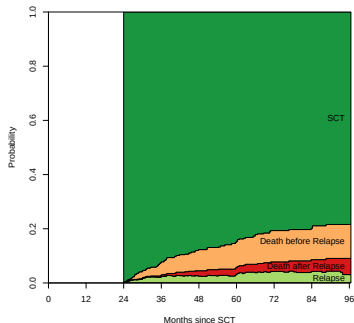
Resulting plots



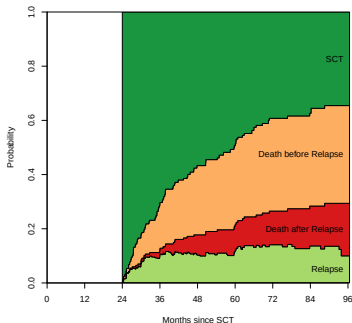
Dynamic forward predictions from $s = 2$ year

```
> ptA2 <- probtrans(msfA, predt=24, direction="forward")
> ptB2 <- probtrans(msfB, predt=24, direction="forward")
```

Good risk patient



Poor risk patient



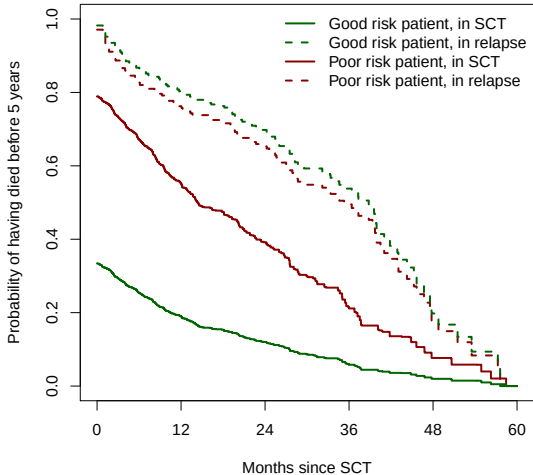
Fixed horizon prediction

- ▶ Often interest is in a fixed prediction horizon, for instance 5 years (60 months)
- ▶ The function *probtrans* can be used with *direction="fixedhorizon"* to obtain dynamic probabilities $P_{kl}(s, t)$ for fixed t and varying s

```
> ptAfh <- probtrans(msfA, predt=60, direction="fixed")  
> ptBfh <- probtrans(msfB, predt=60, direction="fixed")
```

- ▶ When interested in the probability of having died within 60 months post-SCT, we add up `pstate3` (death without relapse) and `pstate4` (death after relapse)

Resulting plot



Other approaches

- ▶ Making models for the transition intensities and subsequently using Aalen-Johansen to obtain predictions is possible, but perhaps somewhat indirect
- ▶ Recent developments try to circumvent making these models and to directly model state or transition probabilities
 - ▶ Pseudo-values (Andersen, Klein)
 - ▶ Direct binomial regression (Scheike)
 - ▶ Landmarking (van Houwelingen)

References



Aalen, O. O., and Johansen, S. (1978).

Empirical transition matrix for nonhomogeneous Markov-chains based on censored observations.

Scand J Statist **5**, 141 – 150.



Andersen, P. K., Borgan, Ø., Gill, R. D., and Keiding, N. (1993).

Statistical Models Based on Counting Processes.

Springer, New York.



Fiocco, M., Putter, H., and van Houwelingen, H. C. (2008).

Reduced-rank proportional hazards regression and simulation-based prediction for multi-state models.

Statist Med **27**, 4340 – 4358.



Putter, H., Fiocco, M., and Geskus, R. B. (2007).

Tutorial in biostatistics: Competing risks and multi-state models.

Statist Med **26**, 2389 – 2430.



de Wreede, L., Fiocco, M., and Putter, H. (2010).

The `mstate` package for estimation and prediction in non- and semi-parametric multi-state and competing risks models.

Comp Meth Prog Biomed **99**, 261–274.